Task-dependent Performance of Single-cell Foundation Models under Low Supervision

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Abstract

Single-cell foundation models are trained on millions of cells, but their value when downstream labels are scarce remains uncertain. Here we compare seven such models with three established representations across clustering, batch correction, cell type annotation, gene expression reconstruction and perturbation prediction. Frozen representations are tested directly for clustering and batch correction and with lightweight task heads for few-shot prediction. CellPLM led the aggregate clustering, annotation and batch-correction comparisons, scMulan led perturbation prediction, and principal component analysis led reconstruction. Of the seven foundation models, one exceeded the strongest classical baseline in clustering, three in batch correction, four in annotation, none in reconstruction and six in perturbation prediction. Representation analyses linked reconstruction performance to preservation of linear gene relationships, but cross-model gene-relation agreement did not exceed a dimension-matched random-projection

null. These results show that the utility of a cell representation depends on the information required by the downstream task.

Introduction

Single-cell technologies measure molecular variation across individual cells and now support reference atlases spanning tissues, developmental stages and disease states [1–4]. These data are high-dimensional, sparse and heterogeneous, so their analysis depends on computational representations that retain relevant variation across cells and genes [5, 6]. Statistical models, machine-learning methods and more recent pretrained models offer different ways to construct such representations.

The information required from a representation varies by task. Cell clustering requires unsupervised recovery of population structure [7], whereas batch correction must remove technical variation without erasing biological heterogeneity [8, 9]. Cell type annotation depends on discriminative features when labelled examples are scarce or query cells originate from another dataset [10–12]. Gene expression reconstruction instead requires quantitative gene relationships, and perturbation prediction requires changes induced by genetic or chemical interventions to remain accessible [13, 14]. An embedding optimized for cell-type separation may therefore support annotation while discarding within-type variation or gene-level covariance needed for reconstruction, trajectory analysis or perturbation-response modelling. A benchmark should test both downstream performance and the biological information retained by the representation [15–17].

Single-cell foundation models (SCFMs) learn reusable representations from large collections of single-cell data. Geneformer [18], scGPT [19], LangCell [20], CellPLM [21], scMulan [22], scFoundation [23], and Nicheformer [24] use transformer-based or related architectures and were pretrained on datasets containing millions of cells. Their representations have been applied to cell type annotation, multimodal integration, trajectory analysis and perturbation prediction [20, 23, 25]. Whether this pretraining consistently produces more useful biological representations than established single-cell methods when downstream labels are scarce remains unclear.

Compared with highly variable gene selection (HVG) [26], Harmony [27], integrated reference workflows [5, 12], UMAP [28], PCA and scVI, SCFMs differ in computational cost, preprocessing, representation dimension and downstream adaptation. Their utility may therefore vary with the biological task, label budget and readout. This variation complicates both method selection and the identification of representation properties that limit transfer under low supervision.

Existing benchmarks cover different parts of this comparison. BioLLM [29] evaluates zero-shot and fine-tuned SCFMs on annotation, drug-response prediction and gene-regulatory analysis; SCMMIB [30] and DANCE [31] emphasize representation learning and multimodal integration; and Wei et al. [32] examine perturbation prediction. Ahlmann-Eltze et al. [33], scSSL-Bench [34], Kedzierska et al. [35] and Wu et al. [36] also show that classical approaches can remain competitive when labels are

limited. These studies motivate a comparison that places multiple biological objectives under one low-supervision protocol and examines the structures retained in the representations alongside downstream scores (Supplementary Note 4 and Supplementary Table 3).

We introduce **CellBench-LS**, which compares seven SCFMs—Geneformer, scGPT, LangCell, CellPLM, scMulan, scFoundation and Nicheformer—with PCA, UMAP and scVI across cell clustering, batch correction, cell type annotation, gene expression reconstruction and perturbation prediction. Clustering and batch correction are evaluated directly from frozen representations without target labels; annotation, reconstruction and perturbation prediction use lightweight task-specific readouts under few-shot supervision. Representation-level diagnostics examine residual cell-state information, held-out gene relationships, perturbation-direction consistency and agreement in embedding-derived gene-relation geometry.

The comparison identifies task-specific rather than model-family-wide gains. CellPLM ranks highly in aggregate clustering, cell type annotation and scale-aligned batch correction; scMulan has the highest perturbation composite, and PCA remains the strongest baseline for gene expression reconstruction. Among the seven SCFMs, one surpasses the strongest classical baseline for clustering, three for batch correction, four for annotation, none for reconstruction and six for perturbation prediction under the current composite metric. The diagnostics provide two boundaries on interpretation. Classical methods better preserve linear co-expression structure, while several SCFMs retain more residual within-type variation. Cross-model agreement in embedding-derived gene-relation geometry does not exceed a dimension-matched random-projection null and therefore cannot by itself support a shared biological mechanism. Representation selection should therefore follow the information required by the downstream analysis rather than the foundation-model label.

Results

We compared SCFMs with classical approaches using common task definitions and downstream readouts (Fig. 1a). Each method receives the corresponding benchmark object, applies its required input transformation and produces a cell-level representation (Fig. 1b). Clustering and batch correction assess frozen representations without cell-type labels. Cell type annotation, gene expression reconstruction and perturbation prediction couple these representations to lightweight task-specific MLP heads under few-shot supervision. The same task-level split rules are applied across methods, although model-specific filtering means that identical cell indices are not guaranteed.

The benchmark spans five tasks that probe distinct properties of single-cell representations (Fig. 1c). Clustering evaluates recovery of cellular population structure, whereas batch correction measures the balance between removal of technical variation and preservation of biological heterogeneity. Cell type annotation tests the discriminative information retained by the embeddings under limited supervision. Gene expression reconstruction examines whether the representations preserve quantitative gene-level information, and perturbation prediction evaluates their ability to capture expression changes induced by genetic or chemical interventions. We further

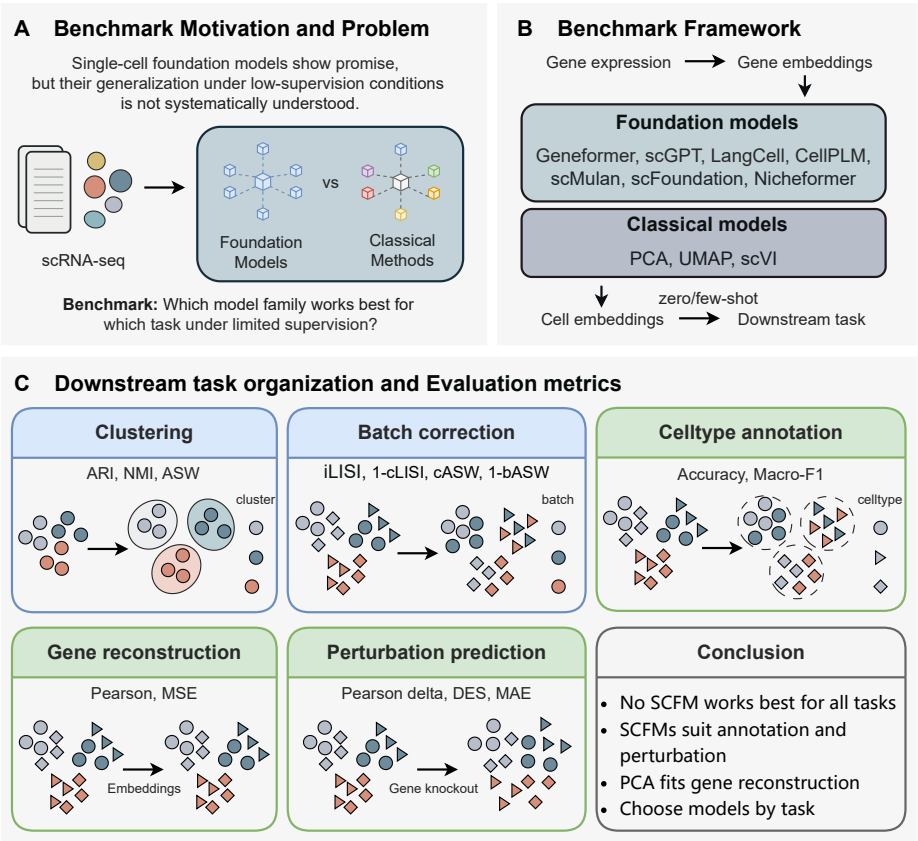


Fig. 1 Overview of the CellBench-LS framework. a, Motivation for systematically evaluating single-cell foundation models (SCFMs) under low-supervision settings. **b**, Unified workflow for generating and evaluating cell representations from representative SCFMs and classical methods. **c**, Evaluation across five core tasks: cell clustering, batch correction, cell type annotation, gene expression reconstruction, and perturbation prediction. Performance varies substantially across models and tasks, with no single SCFM consistently outperforming alternative approaches across all evaluation settings.

complement these task-level evaluations with six representation-level diagnostic analyses designed to determine whether performance is associated with coarse cell-type separation, finer within-cell-type variation, or preservation of gene-relation structure.

Aggregate rankings change across the five tasks (Supplementary Note 5 and Supplementary Table 4). CellPLM achieves the highest scores for clustering (0.6466), cell type annotation (0.5606) and scale-aligned batch correction (AvgBC, 0.5961); scMulan has the highest perturbation composite (0.1114), and PCA has the highest reconstruction score (0.2594). Relative to the strongest classical baseline for each task, one of the seven SCFMs performs better in clustering, three in batch correction, four in cell type annotation, none in gene expression reconstruction and six under the current perturbation composite. The following sections separate these task-level results from the representation analyses used to interpret them.

SCFMs show task- and dataset-dependent performance in clustering and batch correction

Frozen representations produced different clustering and batch-correction rankings across datasets and metrics of annotated populations, geometric separation, batch mixing and biological conservation (Fig. 2).

For clustering, CellPLM achieved the highest aggregate score (AvgCC = 0.6466) and ranked first on six of the nine datasets: PBMC12k, Cortex, Liver, Zheng68k, Lung and Immune (Fig. 2a; Supplementary Table 5). Its highest scores occurred on PBMC12k (0.8062) and Lung (0.7912), where ARI and NMI agreement with the reference labels was accompanied by a high scaled silhouette score. Other models led on the remaining datasets: PCA on hPancreas (0.6968), scVI on MS (0.5856) and Nicheformer on Covid (0.5249). Metric-level rankings also differed. UMAP, for example, had the highest scaled silhouette score on hPancreas (0.7696), while PCA had the higher aggregate score because of its ARI and NMI. The two-dimensional projections likewise show that compact islands can coexist with fragmentation of annotated populations (Fig. 2b). CellPLM is the only SCFM that exceeds the strongest classical baseline in the aggregate clustering comparison.

Batch-correction metrics exposed a trade-off between batch mixing and biological conservation (Fig. 2c,d; Supplementary Table 6). On UC-EPI, Nicheformer had the highest raw iLISI (13.44), while CellPLM had the highest cASW (0.5394). UMAP had the highest $1 - \text{bASW}$ (0.6272) and the least negative stored $1 - \text{cLISI}$ value (-0.58), despite a lower raw iLISI (6.69). After scale alignment and aggregation of iLISI, cell-type LISI, cASW and $1 - \text{bASW}$ across UC-IMM and UC-EPI, CellPLM ranked first (AvgBC = 0.5961), followed by Nicheformer (0.5935), scFoundation (0.5853) and scVI (0.5818). The epithelial-cell projections show that the models retain major epithelial states through different geometries (Fig. 2d): CellPLM preserves clearer state organization, whereas Nicheformer emphasizes batch mixing.

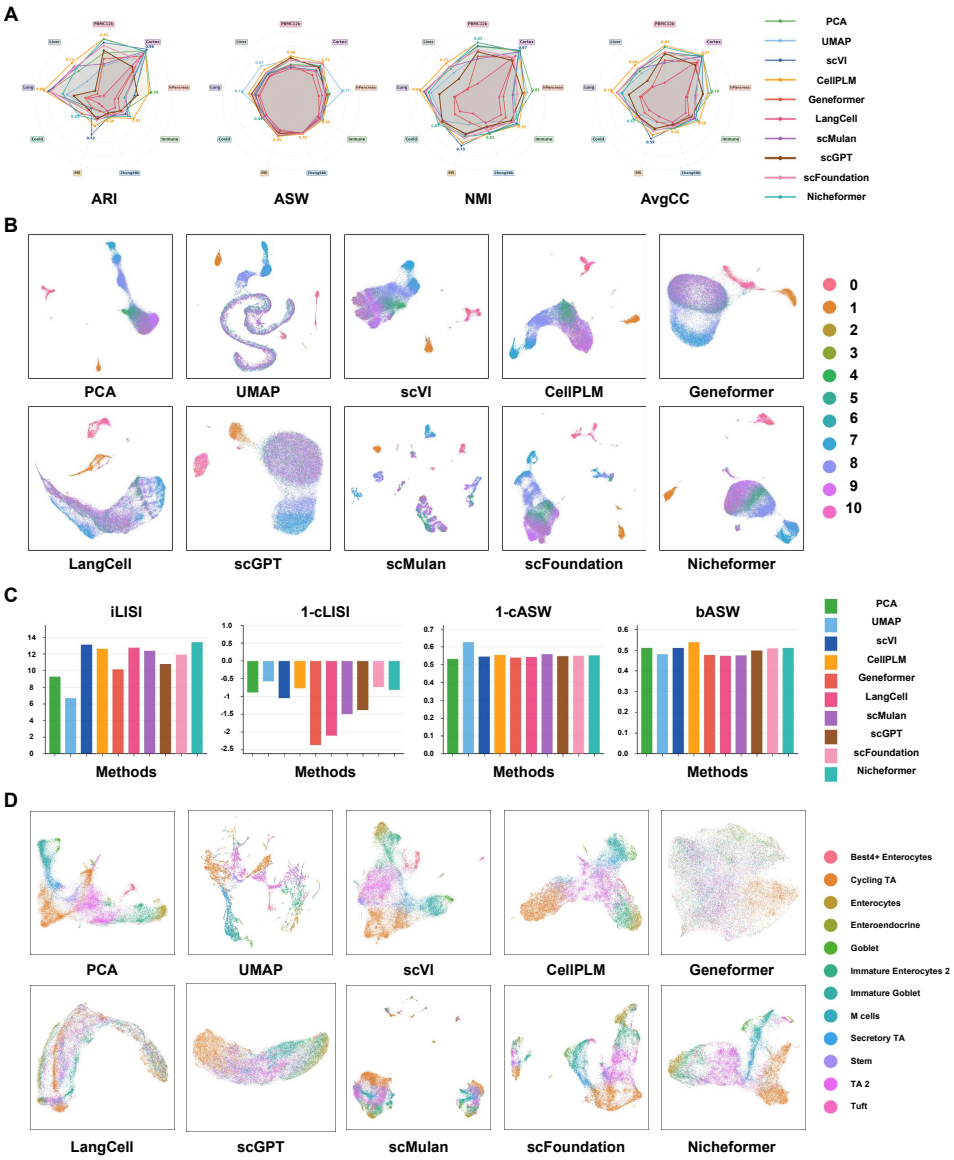


Fig. 2 Clustering and batch-correction performance of frozen cell representations. a, Clustering performance of ten methods across nine datasets. Radar plots summarize ARI, NMI, the scaled silhouette score $\mathrm{ASW}_n = (\mathrm{ASW} + 1)/2$, and AvgCC, with higher values indicating better performance. **b**, Representative two-dimensional projections of frozen embeddings generated by the ten methods, coloured by reference cell labels. **c**, Batch-correction performance on UC-EPI evaluated using raw iLISI, the stored pre-normalization 1-cLISI value, 1-bASW, and cASW across the ten methods. In the current source figure, the axes labelled "1-cASW" and "bASW" display 1-bASW and cASW, respectively; all text and aggregate calculations use the latter metric definitions. **d**, Two-dimensional projections of the corrected UC-EPI epithelial-cell representations, coloured by reference epithelial cell type. ARI, adjusted Rand index; NMI, normalized mutual information; ASW, average silhouette width; LISI, local inverse Simpson index; cASW, cell-type ASW; bASW, batch ASW.

Few-shot annotation reveals uneven label efficiency across pretrained representations

Few-shot annotation tests both performance with very sparse labels and improvement as the label budget increases. These two properties did not produce the same model ranking across the nine evaluated datasets; Fig. 3a displays eight, and the complete results are reported in Supplementary Tables 7 and 8.

CellPLM combined high accuracy under minimal supervision with gains as additional labels were introduced (Supplementary Tables 7 and 8). On PBMC12k, its accuracy increased from 0.818 in the 1-shot setting to 0.930 in the 9-shot setting; on MS, it increased from 0.528 to 0.693. scVI showed a larger gain on MS, from 0.309 to 0.687, and nearly matched CellPLM at 9-shot despite its lower 1-shot accuracy. The learning curves also varied by dataset. Cortex was comparatively easy, with scFoundation and Nicheformer exceeding 0.97 accuracy at 1-shot and approaching 0.99 at 9-shot, while Zheng68k and MS remained more difficult. On hPancreas, UMAP started at 0.651, whereas scFoundation increased from 0.475 to 0.936 as the label budget grew. In the 1-shot summary averaged across all nine datasets, CellPLM achieved the highest annotation score (AvgCA = 0.5606), followed by Nicheformer (0.5213), scFoundation (0.5150) and scMulan (0.4799); UMAP reached 0.4572. The curves distinguish representations that perform well from a single labelled cell from those that require a larger label budget.

The Immune dataset projections provide a visual comparison of the same representations (Fig. 3b). Several SCFMs organize major immune populations into more distinct regions, but two-dimensional separation does not map directly to classification accuracy. Overlapping projected populations can remain discriminative in the full representation, while separated regions can reflect distortion introduced by dimensionality reduction. The annotation comparison therefore relies on the supervised metrics rather than on the projections alone.

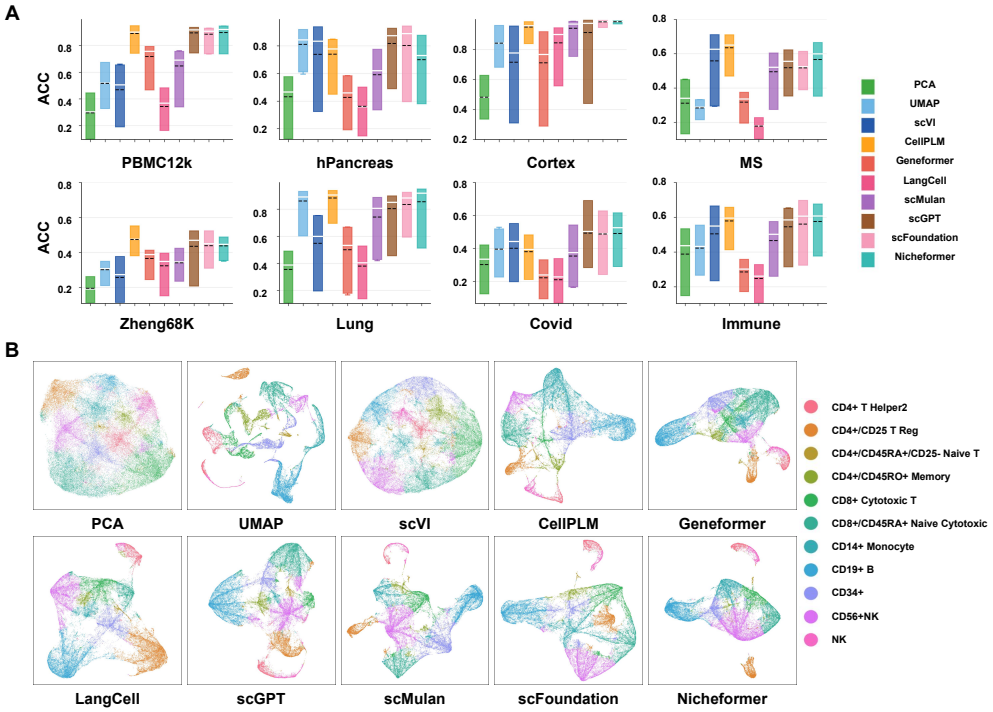


Fig. 3 Few-shot cell type annotation. a, Few-shot annotation accuracy (ACC) of ten frozen representations across eight of the nine benchmark datasets using the same lightweight classifier; results for Liver are reported in Supplementary Table 7. For each method, the coloured vertical interval summarizes performance across $k \in \{1, 3, 5, 7, 9\}$ labelled examples per cell type; the lower and upper boundaries correspond to the 1-shot and 9-shot settings, respectively. Dashed lines indicate the mean accuracy and white lines indicate the median accuracy across few-shot settings. Results are averaged over five random seeds. **b**, Two-dimensional visualizations of the frozen representations for the Immune dataset, with cells coloured by annotated cell type. The panels illustrate differences in representation geometry and cell-type organization across classical methods and SCFMs. ACC, accuracy; SCFM, single-cell foundation model.

Transcriptional readout tasks reveal model- and metric-dependent performance

Gene reconstruction and perturbation prediction test information that is not captured by cell-identity metrics alone. We evaluated reconstruction across nine datasets and perturbation prediction on Adamson and Norman; Fig. 4 displays eight reconstruction datasets, with Liver reported in Supplementary Tables 9 and 10.

PCA led reconstruction on most datasets, with the highest mean Pearson correlation and lowest mean MSE on hPancreas, Cortex, Liver, Zheng68k, Lung, Covid and Immune (Fig. 4a; Supplementary Tables 9 and 10). scMulan had the highest mean Pearson and lowest mean MSE on PBMC12k and MS. The $k = 100$ aggregate across all nine datasets consequently ranked PCA first (AvgGR = 0.2594). On Immune, the ordering by reconstruction did not match the visual separation of annotated cell

types (Fig. 4b), illustrating that the two readouts measure different representation properties.

Perturbation rankings differed across PDS, MAE, differential-expression score (DES) and AvgPP (Fig. 4c; Supplementary Tables 11 and 12). scFoundation had comparatively low MAE but greater variability in DES. CellPLM, Geneformer, LangCell and scMulan had similar central DES values despite differences in perturbation discrimination and expression error, leaving modest AvgPP differences among several high-ranked representations. The two-dimensional projections also differed in the separation of control and perturbed cells (Fig. 4d). Model ranking therefore depends on whether evaluation emphasizes identification of the perturbation effect, expression error or differential-expression recovery.

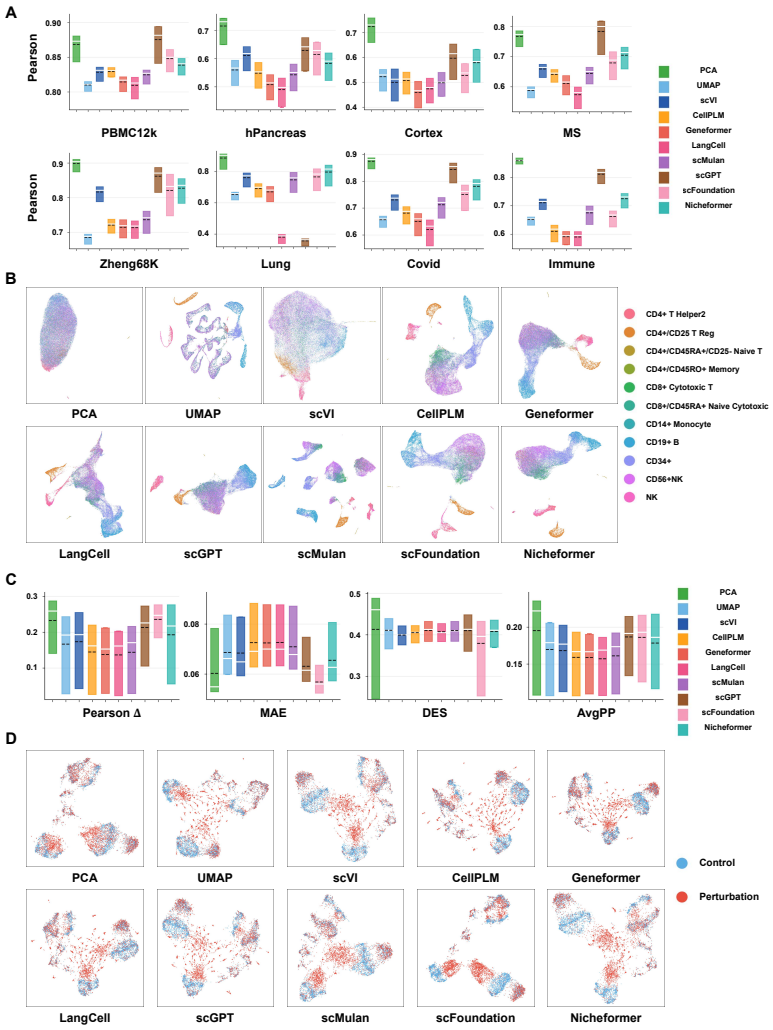


Fig. 4 Gene-expression reconstruction and perturbation prediction. **a**, Gene-expression reconstruction performance of ten frozen representations across eight of the nine benchmark datasets, evaluated by Pearson correlation for the top 400 highly variable genes identified using Scanpy; Liver results and MSE values are reported in Supplementary Tables 9 and 10. For each method, vertical intervals summarize performance across $k \in \{100, 300, 500, 700, 900\}$ training cells per cell type; dashed lines indicate the mean and white lines indicate the median across the evaluated training-set sizes. **b**, Two-dimensional visualizations of the frozen representations for the Immune dataset, with cells coloured by annotated cell type. **c**, Perturbation-prediction performance evaluated using perturbation discrimination score (PDS), MAE, DES and AvgPP. Vertical intervals summarize performance across $k \in \{1, 3, 5, 7, 9\}$ labelled cell pairs per seen perturbation condition. Dashed lines indicate the mean and white lines indicate the median across few-shot settings. The axis labelled "Pearson Δ" in the current source figure contains the PDS values reported here and in Supplementary Table 12. **d**, Two-dimensional visualizations of the frozen representations for perturbation-prediction data, with cells coloured by control or perturbation condition. MSE, mean squared error; PDS, perturbation discrimination score; MAE, mean absolute error; DES, differential expression score; AvgPP, average perturbation-prediction composite.

Representation-level diagnostics

Six diagnostic analyses examine cell-level, gene-level and perturbation-associated structure in the embeddings (Fig. 5; Supplementary Note 6). They test variation beyond coarse cell-type identity, preservation of expression and perturbation-response structure, and agreement between independently trained models.

Within annotated cell types, chance-adjusted kNN overlap exceeded a 200-permutation null for every model–dataset combination across five atlas datasets (BH-FDR $q = 0.005$, the empirical floor of $1/201$; Fig. 5a; Supplementary Table 13). The effect size varied by representation. PCA achieved the highest mean adjusted kNN overlap (0.536), while scFoundation achieved the highest fold-internal expression-PC R^2 (0.282). The two readouts gave different rankings: CellPLM, for example, retained relatively high expression-PC predictability ($R^2 = 0.246$) despite a lower adjusted neighborhood overlap (0.166). Local neighborhoods and continuous expression axes thus capture different aspects of within-type structure.

Subtracting training-cell estimates of cell-type centroids from both embedding and expression spaces imposed a stricter test (Fig. 5e; Supplementary Table 14). Expression-PC R^2 decreased for every model after residualization, while chance-adjusted kNN overlap was retained or increased for all ten methods. Post-residualization overlap exceeded both a within-type permutation null and a dimension-matched random-projection null for every model and dataset (200 draws per null; BH-FDR $q = 0.005$ throughout). PCA had the highest post-residualization adjusted overlap (0.709) and expression-PC R^2 (0.654). Its adjusted overlap increased from 0.565 to 0.709 while PC R^2 decreased from 0.941 to 0.654; Geneformer changed from 0.122 to 0.137 and from 0.643 to 0.116, respectively. Local cellular organization therefore persists after removal of cell-type-level offsets, although much of the PC-predictive variation does not.

Held-out gene–gene relationships provide a direct test of information relevant to reconstruction. Across seven compatible datasets, PCA achieved the highest mean held-out Spearman (0.491) and Pearson (0.636) correlations among the ten representations (Fig. 5b; Supplementary Table 15). The next-best values were scVI at 0.522 for Pearson and Nicheformer at 0.466 for Spearman. Gene loadings were fitted on training cells and evaluated on disjoint validation cells, excluding same-cell fitting as an explanation. PCA's advantage, especially for Pearson correlation, links its reconstruction performance to preservation of magnitude-sensitive linear covariance.

Perturbation identity remained linearly accessible without overt geometric separation. On Adamson and Norman, fold-internal multinomial probes achieved macro-F1 above a 200-permutation label-shuffling null for every model (Fig. 5d), although perturbation ASW remained negative across all methods (-0.329 to -0.007; Supplementary Table 16). scVI achieved the highest macro-F1 and balanced accuracy on both datasets, with ASW of -0.063 on Adamson and -0.007 on Norman. Perturbation prediction can therefore remain feasible when control and perturbed cells overlap under cluster-oriented diagnostics.

Pathway consistency imposed a stronger criterion than perturbation decoding. For single-gene perturbations in Norman, LangCell produced the largest within- versus between-pathway Δ cosine (0.254), followed by scVI (0.225; Fig. 5c; Supplementary

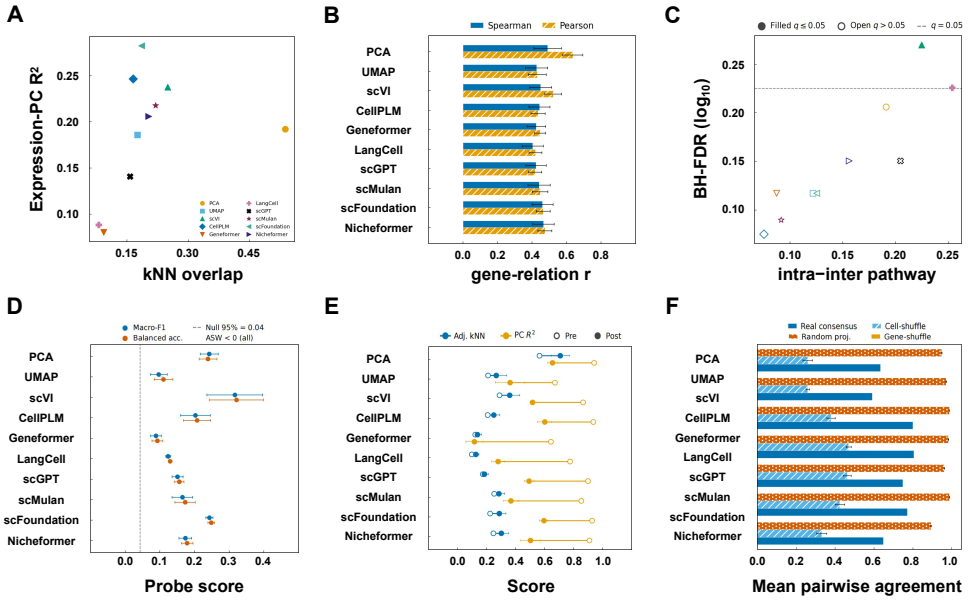


Fig. 5 Mechanistic representation diagnostics across six complementary tests. a, Test 2 quantifies within-cell-type structure across five atlas datasets by comparing chance-adjusted kNN overlap with fold-internal expression-PC R^2. Adjusted kNN overlap is evaluated against within-type permutation nulls, whereas PC R^2 is reported as an effect-size measure. **b**, Test 4 evaluates held-out gene-relation preservation across seven compatible datasets using Spearman and Pearson correlations, with gene loadings estimated exclusively from training cells. **c**, Test 5 assesses whether single-gene perturbation directions on Norman align with pathway structure. The vertical axis reports $-\log_{10}(q)$ for BH-FDR-adjusted q values from a 200-permutation gene-label null, so larger values indicate stronger evidence. Filled symbols denote $q \leq 0.05$, and the dashed line is drawn at $-\log_{10}(0.05) = 1.301$. **d**, Test 1 measures perturbation-associated information on Adamson and Norman using fold-internal macro-F1 and balanced accuracy. Error bars summarize variation across folds, the dashed reference denotes the 95th percentile of the 200-permutation label-shuffling null, and perturbation ASW is shown as an auxiliary measure. **e**, Test 3 compares representations before and after removal of cell-type centroids from both embedding and expression spaces. Paired values show that adjusted kNN overlap is largely retained or increased after residualization, whereas expression-PC R^2 generally decreases. **f**, Test 6 reports dataset-level cross-model consensus for seven datasets. For each dataset, real consensus is the mean Pearson agreement between embedding-derived gene-relation matrices over all $\binom{10}{2} = 45$ model pairs, averaged across gene-set repeats, and is compared with cell-shuffling, gene-shuffling, and dimension-matched random-projection nulls. Real consensus exceeds the shuffle-based nulls but remains below the random-projection baseline in every dataset. kNN, k-nearest neighbour; PC, principal component; BH-FDR, Benjamini–Hochberg false discovery rate.

Table 17). After BH-FDR correction across ten models, scVI remained significant ($q = 0.033$), LangCell was at the threshold ($q = 0.050$), PCA fell just beyond it ($q = 0.060$), and the other seven models had $q > 0.09$. Panel c reports $-\log_{10}(q)$, placing $q = 0.033$ at 1.48 and the $q = 0.05$ threshold at 1.30. Decodable perturbation identity therefore does not establish pathway-level organization.

Cross-model gene-relation geometry was evaluated at the dataset level (Fig. 5f; Supplementary Table 18). For each of seven datasets, consensus is the mean of 45 pairwise Pearson correlations among the ten model-specific relation matrices, averaged across gene-set repeats. Consensus exceeded the shuffle-cell and shuffle-gene controls

in every dataset and ranged from 0.591 to 0.805; shuffle-gene agreement remained near zero (−0.0010 to 0.0008). Dimension-matched random projections produced higher agreement, however, with means of 0.894–0.988 in every dataset. The real-minus-random deficit ranged from −0.177 in Cortex to −0.380 in Adamson. The observed agreement is therefore compatible with shared expression geometry and does not establish convergence on a common regulatory mechanism.

Discussion

Large-scale pretraining did not produce a consistent advantage across the five low-supervision tasks. CellPLM ranked highest in clustering, few-shot annotation and scale-aligned batch correction, while scMulan had the highest perturbation composite. PCA remained the strongest aggregate reconstruction baseline, and scVI was competitive in several cellular and perturbation analyses. SCFMs therefore extend the available set of single-cell representations, but their value depends on the downstream task [15, 16, 35, 36].

The representation analyses identify why some rankings diverge. Within-type neighborhood structure persisted after cell-type centroids were removed, so clustering and annotation scores cannot be attributed only to coarse cell-type separation. Residualization nevertheless reduced expression-PC predictability in every model. Local neighborhood preservation and recovery of dominant expression axes therefore measure different information. This difference matters for reconstruction: PCA best preserved held-out linear gene–gene relationships, particularly under Pearson correlation, despite not leading the cell-identity tasks. A representation can separate cell types without retaining the covariance structure required by a gene-level readout.

Perturbation identity was linearly decodable above shuffled-label nulls for every representation on Adamson and Norman, although perturbation silhouette scores remained negative. Decodability therefore does not require compact perturbation clusters. Pathway organization was less common: after multiple-testing correction, only a minority of models showed supported pathway-consistent perturbation geometry. These analyses separate access to perturbation identity from geometric separation and pathway structure. They also define the scope of the prediction benchmark. The few-shot protocol evaluates perturbation conditions represented during training and does not test generalization to unseen perturbations [13, 14, 32, 33].

Cross-model agreement has a similar interpretive boundary. Embedding-derived gene-relation geometries agreed more than the shuffle-cell and shuffle-gene controls, but not more than a dimension-matched random-projection null in any of the seven datasets. Generic projections of the same expression structure can therefore account for part of the observed agreement. Curated pathways or transcription-factor–target relationships would be needed to test whether the shared geometry contains regulatory information [37]. Other design choices also limit the rankings. Batch-correction scores measure each representation followed by PCA→Harmony, not native batch invariance, and native embedding dimension or layer selection may affect the downstream probes. Matching probe capacity, dimensionality and layer selection would help isolate these effects.

The aggregate rankings are descriptive rather than inferential. AvgGR and AvgPP combine components on their native scales, so they are convenient summaries of this protocol but are not invariant to rescaling; the constituent metrics remain the primary evidence. Five random seeds capture variation in the supervised readouts, but no paired dataset-level model comparison is reported, and batch correction includes only two datasets. In addition, model-specific preprocessing and filtering mean that common split rules do not guarantee identical cell indices across representations. Prespecified shared splits, uncertainty intervals and a normalized-score sensitivity analysis would provide a stricter comparison.

Model scale alone does not specify which biological information a representation will preserve. Objectives that improve cell-state discrimination may alter gene-level covariance, while expression-preserving representations may still lack pathway organization of perturbation responses. Future evaluations should report these properties separately and compare biological geometry with matched null models. CellBench-LS provides such comparisons under a common low-supervision protocol and supports choosing representations for the downstream readout rather than for pretraining scale alone.

Methods

Benchmark design

Single-cell foundation models (SCFMs) differ in architecture, tokenization and pretraining objective, but each produces a cell-level representation. **CellBench-LS** compares these representations with classical methods using common task definitions, split rules and downstream readouts while retaining the input processing required by each model (Fig. 1). Because filtering occurs within model-specific pipelines, the rules are shared but the retained cell indices need not be identical. Task-level metrics quantify downstream utility, while representation-level analyses examine the biological structure retained in the embeddings.

Preprocessing and embedding extraction. CellBench-LS includes diverse scRNA-seq datasets spanning general representation learning, batch correction and perturbation-response analysis. Dataset-specific characteristics and experimental settings are described in the Dataset description subsection, Supplementary Note 1 and Supplementary Table 1.

Let $\mathbf{X} \in \mathbb{R}^{N \times G}$ denote the single-cell expression matrix for N cells and G genes. Dataset-specific loading and quality-control steps define the benchmark expression matrix. Depending on the dataset and model requirements, expression values are normalized or log-transformed and highly variable genes are identified before representation extraction. Each cell profile $\mathbf{x}_i \in \mathbb{R}^G$ is then processed using the model-specific input pipeline, including, where required, binning, tokenization or gene ranking, to obtain a latent representation $\mathbf{z}_i \in \mathbb{R}^{d_i}$. Representation dimensionality d_i is model specific.

Models. We evaluated seven representative SCFMs—scGPT, Geneformer, LangCell, CellPLM, scMulan, scFoundation and Nicheformer—together with three classical baselines: PCA, UMAP and scVI. Unless otherwise stated, SCFM representations were

generated using the official pretrained checkpoints and their corresponding preprocessing procedures. Additional model details are provided in the Model Descriptions subsection.

Downstream evaluation. The benchmark comprises five tasks representing distinct requirements of single-cell representations: cell clustering, batch correction, cell type annotation, gene expression reconstruction and perturbation prediction. All evaluations use fixed cell representations $\mathbf{z}_i$. Unsupervised tasks are assessed directly from frozen representations, whereas supervised tasks use standardized lightweight MLP heads under controlled few-shot settings. This design limits task-specific adaptation; observed differences nevertheless reflect both the extracted representations and their required model-specific preprocessing.

Cell clustering. Cell clustering evaluates whether frozen representations recover cellular population structure without target-label supervision. Louvain clustering [38] is applied directly to the extracted representations, and agreement with reference annotations is quantified using adjusted Rand index (ARI), normalized mutual information (NMI) and average silhouette width (ASW). For reporting and aggregation, the raw silhouette score is mapped from $[-1, 1]$ to $[0, 1]$ as $\mathrm{ASW}_n = (\mathrm{ASW} + 1)/2$.

Batch correction. Batch correction evaluates the extent to which a representation supports removal of technical variation while preserving biological structure [8, 16]. PCA followed by Harmony is applied to each frozen representation before evaluation using iLISI, cell-type LISI, cASW and 1−bASW. If s_{cell} and s_{batch} denote the corresponding raw silhouette scores, the reported ASW terms are $\mathrm{cASW} = (s_{\mathrm{cell}}+1)/2$ and $1 - \mathrm{bASW} = (1 - s_{\mathrm{batch}})/2$. The resulting scores therefore characterize the performance of the representation together with the standardized downstream correction procedure rather than the intrinsic batch invariance of the frozen representation alone.

Cell type annotation. Cell type annotation evaluates the discriminative information retained by each representation under limited supervision. For each random seed, cells are randomly divided into an 80% training pool and a 20% held-out test set. An MLP classifier maps each cell representation $\mathbf{z}_i$ to its corresponding label y_i. From the training pool, we sample $k \in \{1, 3, 5, 7, 9\}$ labelled cells per cell type without replacement; the remaining training-pool cells are not used to fit the classifier. Performance on the held-out test set is assessed using accuracy, macro-F1, precision and recall.

Gene expression reconstruction. Gene expression reconstruction evaluates the extent to which cell representations retain quantitative gene-level information. For each random seed, cells are randomly divided into 80% training, 10% validation and 10% test partitions. A regression head predicts expression values for the top 400 highly variable genes identified using Scanpy from each cell representation. From the training partition, we sample $k \in \{100, 300, 500, 700, 900\}$ cells per cell type without replacement. The validation partition is used for checkpoint selection, and reconstruction performance is assessed on the held-out test partition using mean squared error (MSE) and Pearson correlation.

Perturbation prediction. Perturbation prediction evaluates whether cell representations support prediction of transcriptional responses under limited supervision for perturbation conditions observed during training. A fixed 10% of perturbed cells,

selected with random seed 42, forms the test partition. The remaining perturbed cells are divided into training and validation partitions in an 8:1 ratio for each experimental seed. For each example, the control-cell representation $\mathbf{z}_{\text{ctrl}}$ is concatenated with a perturbation-identity embedding g_p, and an MLP predicts the perturbation-induced expression residual Δ. The predicted expression profile is $\hat{\mathbf{x}} = \mathbf{x}_{\text{ctrl}} + \Delta$. From the training partition, we sample $k \in \{1, 3, 5, 7, 9\}$ labelled examples per observed perturbation condition. The validation partition is used for checkpoint selection. Performance on the fixed test partition is evaluated using perturbation discrimination score (PDS), Differential Expression Score (DES) and mean absolute error (MAE), with AvgPP reported as a compact composite summary. This protocol assesses low-supervision prediction for seen perturbation conditions and is not intended as an evaluation of out-of-distribution generalization to unseen perturbations.

Model descriptions

All methods are evaluated through a common embedding interface. The methods differ in the information and objectives used to construct their cell representations.

PCA. Principal component analysis projects gene-expression profiles onto orthogonal axes that capture the largest sources of variance and serves as the linear baseline [39].

UMAP. Uniform manifold approximation and projection maps high-dimensional observations to a lower-dimensional space while preserving local neighborhood structure and serves as the nonlinear geometric baseline [40].

scVI. scVI is a variational autoencoder for single-cell transcriptomics that learns probabilistic latent representations while modelling count variability and technical effects, including batch variation [41].

Geneformer. Geneformer represents each cell as a rank-ordered sequence of expressed genes and uses self-supervised transformer pretraining to learn context-dependent gene relationships [18].

scGPT. scGPT is a generative transformer for single-cell and multi-omic data that jointly models gene identities and expression values through self-supervised sequence learning [19].

LangCell. LangCell aligns single-cell transcriptomes with natural-language descriptions of cell identity in a shared embedding space and therefore introduces textual supervision into pretraining [20].

CellPLM. CellPLM represents cells as tokens and tissues as higher-order contextual sequences, incorporating intercellular context during pretraining [21].

scMulan. scMulan represents single-cell expression profiles as structured sequences and jointly optimizes multiple generative pretraining objectives [22].

scFoundation. scFoundation is pretrained on large human single-cell transcriptomic collections to generate transferable cell representations [23].

Nicheformer. Nicheformer is pretrained on single-cell and spatial transcriptomic data and incorporates spatial context alongside cell-intrinsic transcriptional state [24].

Dataset description

To assess model performance across diverse biological and experimental settings, we use a collection of publicly available single-cell datasets covering general representation learning, batch correction, and perturbation prediction. Summary statistics and source accessions are provided below, in Supplementary Note 1 and in Supplementary Table 1; task-specific evaluation settings are described in the corresponding Methods subsections.

PBMC12k. PBMC12k comprises 11,990 peripheral blood mononuclear cells from a healthy donor, with nine benchmark labels spanning major immune populations, including B cells, $CD4^+$ and $CD8^+$ T cells, NK cells, monocytes, and dendritic cells. We use this dataset to evaluate representations of closely related immune cell states. Data are available from 10x Genomics [42].

hPancreas. hPancreas is an integrated human pancreas dataset assembled from multiple studies and experimental batches. The source object contains 16,382 cells; after benchmark-specific quality control and filtering, 14,818 cells, 3,000 genes, and 14 cell-type labels are retained for embedding extraction and evaluation. The dataset contains both endocrine and exocrine populations and is used to assess representation transfer across studies. Source datasets are listed in the Hemberg laboratory single-cell dataset collection [16].

MS. The MS dataset is derived from the human multiple sclerosis scRNA-seq study of Schirmer et al. and comprises samples from nine healthy controls and 12 individuals with MS. The benchmark cohort contains 21,312 cells, of which 7,844 control cells form the reference set for model fine-tuning and 13,468 MS cells form the out-of-distribution query set. The corresponding summary statistics in Supplementary Table 1 refer to the combined cohort. Cell types observed exclusively in the query set—B cells, T cells, and oligodendrocyte B—are excluded from the corresponding transfer evaluation. Original cell annotations are used as ground truth, and 3,000 highly variable genes (HVGs) are retained. Data are available from E-HCAD-35 [43].

Zheng68k. Zheng68k is derived from the 10x Genomics Fresh 68k PBMCs (Donor A) dataset, which contains 68,579 cells; the processed benchmark object retains 68,450 cells assigned to 11 annotated cell types. Its relatively large sample size and diverse immune populations provide a complementary setting for evaluating cell representation and classification at scale. Source data are available from 10x Genomics [2].

Cortex. The Cortex dataset is derived from the Perirhinal Cortex subset of the adult human brain single-nucleus transcriptomic atlas of Siletti et al. (2023) [44]. The benchmark subset comprises two batches containing 8,465 and 9,070 nuclei, respectively, yielding 17,535 nuclei and 59,236 measured genes in total. We use this dataset for label-free, few-shot, and full-data integration analyses. The processed resource used by the benchmark is available from the scVI reproducibility repository.

Covid. The Covid dataset comprises 274,346 cells distributed across 18 batches and includes cells derived from lung tissue, peripheral blood, and bone marrow. The benchmark uses all 274,346 cells, 18,474 genes, and 39 annotated cell populations;

no additional 20,000-cell subsampling is applied during embedding extraction or evaluation. Original annotations are retained throughout. Data are available from the scArches reproducibility repository [12].

Liver. The Liver dataset contains 56,721 single-cell transcriptomes from 46 hepatocellular carcinoma (HCC) and intrahepatic cholangiocarcinoma (iCCA) biopsies collected from 37 patients. It provides a heterogeneous tumour microenvironment setting for evaluating representations across malignant and non-malignant cell populations. Data are available from GEO under accession GSE151530 [45].

Lung. The Lung dataset comprises 208,506 cells from 58 lung adenocarcinoma samples obtained from 44 patients. Samples span primary tumours, lymph-node and brain metastases, pleural effusions, and matched normal lung and lymph-node tissues, providing substantial biological and anatomical heterogeneity. Data are available from GEO under accession GSE131907 [46].

Immune. The Immune dataset comprises 329,762 cells from immune compartments across 16 tissues obtained from 12 adult donors and profiled using single-cell RNA sequencing together with V(D)J sequencing. We use the transcriptomic profiles to evaluate representation quality across diverse tissue-resident and circulating immune populations. Data are available from ArrayExpress/BioStudies under accession E-MTAB-11536 [47].

UC-EPI and UC-IMM. The benchmark object is derived from the ulcerative-colitis colon-mucosa atlas of Smillie et al. [48] and distributed with the scPhere study [49]. It contains 301,749 cells from 68 biopsies collected from 18 individuals with UC and 12 healthy controls and profiled using 10x Chromium v1 and v2 chemistries. Cells are assigned to three major compartments: stromal/glial (26,678 cells), epithelial (64,457 cells), and immune (210,614 cells), with further subtype annotation based on unsupervised clustering and manual curation. For batch-correction experiments, we use the epithelial (UC-EPI) and immune (UC-IMM) compartments and retain the 1,361 and 1,068 HVGs, respectively, provided in the processed objects. These processed data are available from the Single Cell Portal under accession SCP551.

Adamson. Adamson is a CRISPR-based perturbation dataset comprising 68,603 K562 cells, 5,060 genes, and 86 perturbation conditions. We use it to evaluate the extent to which learned representations capture transcriptional responses to genetic perturbations. Source data are available from GEO under accession GSE90546 [50].

Norman. Norman comprises 91,205 single cells, 5,045 genes, and 283 perturbation conditions generated using CRISPR-based genetic perturbations. Compared with Adamson, its broader perturbation space provides a more challenging setting for evaluating representations of perturbation-induced transcriptional states. Data are available from GEO under accession GSE133344 [51].

Downstream head architecture and training

For all few-shot downstream tasks, we employed lightweight task-specific multilayer perceptron (MLP) heads trained on top of frozen embeddings, while keeping the pretrained representation backbone unchanged. Each MLP head consisted of fully connected layers with nonlinear activation functions, batch normalization, and dropout

regularization. The output layer was adapted for classification, reconstruction, or perturbation prediction. Annotation heads were trained for 500 epochs. Reconstruction and perturbation heads were trained for at most 500 epochs and evaluated on a validation split every 20 epochs; training stopped after five consecutive validation checks without an MSE improvement greater than 10^{-4}, and the best validation checkpoint was restored.

For cell type annotation and gene expression reconstruction, the MLP heads were optimized using a learning rate of 1×10^{-3} with a batch size of 32 for a maximum of 500 epochs. For perturbation prediction, a higher learning rate of 5×10^{-3} was used with the same batch size and training duration. Cell clustering and batch correction were evaluated directly using the frozen embeddings without additional task-specific model training. Supplementary Note 4 and Supplementary Table 2 list the task-specific settings.

Evaluation metrics

Model performance was evaluated using task-specific metrics together with compact composite scores designed for consistent comparison across downstream analyses. Unless otherwise specified, all composite scores were formulated such that higher values indicate better performance. The composite scores used throughout the main text were defined as:

$$\mathrm{AvgCC} = \frac{1}{3}(\mathrm{ARI} + \mathrm{NMI} + \mathrm{ASW}_n) \tag{1}$$

where AvgCC summarizes cell clustering performance by integrating clustering agreement with reference annotations and intrinsic cluster separation.

$$\mathrm{AvgBC} = \frac{1}{4}(\mathrm{iLISI}_n + \text{1-}\mathrm{cLISI}_n + \mathrm{cASW} + \text{1-}\mathrm{bASW}) \tag{2}$$

where AvgBC evaluates batch correction by jointly considering batch mixing and preservation of biological structure. The normalized components combine local batch diversity, cell-type conservation, and batch-associated separation.

$$\mathrm{AvgCA} = \frac{1}{2}(\mathrm{Acc} + \mathrm{F1}) \tag{3}$$

where AvgCA measures cell type annotation performance by combining overall classification accuracy and macro-averaged F1 score.

$$\mathrm{AvgGR} = \frac{1}{2}(\mathrm{Pearson} - \mathrm{MSE}) \tag{4}$$

where AvgGR summarizes gene expression reconstruction performance by balancing expression pattern preservation and reconstruction error.

$$\mathrm{AvgPP} = \frac{1}{3}(\mathrm{PDS} + \mathrm{DES} - \mathrm{MAE}) \tag{5}$$

where AvgPP integrates perturbation response prediction accuracy, recovery of differentially expressed genes, and prediction error.

These composite scores are used for concise visualization and descriptive ranking. Individual task-specific metrics remain the primary basis for biological interpretation. AvgGR and AvgPP combine components retained on their native scales and are therefore specific to the present evaluation protocol rather than invariant to rescaling. Their rankings should be read alongside the corresponding Pearson, MSE, PDS, DES and MAE values. Supplementary Note 2 provides the full metric definitions, normalization procedures and implementation details.

Representation-level analyses

Task-level performance alone does not establish whether a model captures fine-grained cellular states or primarily reflects coarse differences between cell types. We therefore complemented CellBench-LS with six representational analyses of perturbation-specific, within-cell-type, gene-level and cross-model structure in the frozen embeddings. Preprocessing and fitted transformations were estimated from training data where applicable. Inferential comparisons use the null models and multiple-testing corrections specified for each analysis; quantities without a corresponding test are reported as effect sizes. Supplementary Note 3 provides dataset eligibility, null construction and implementation details.

For model m, let $\mathbf{z}_i^{(m)} \in \mathbb{R}^{d_m}$ denote the frozen embedding of cell i, $\mathbf{x}_i$ its expression profile, and c_i its cell-type or perturbation label, as appropriate.

Perturbation-associated information. We first assessed whether the embeddings retain perturbation-specific information beyond coarse cellular identity. A linear multinomial probe was trained to predict perturbation conditions from frozen embeddings, and performance was evaluated on held-out cells using macro-F1 and balanced accuracy. Perturbation-associated separation in embedding space was additionally quantified using perturbation silhouette width. Performance exceeding a label-permutation null indicates that perturbation-related information remains linearly accessible from the representation.

Within-cell-type structure. We next tested whether embeddings preserve local cellular organization after explicitly conditioning on cell type. For each cell, we compared its within-cell-type nearest-neighbor sets in expression and embedding space and defined the chance-adjusted within-type neighborhood fidelity as

$$\mathrm{WNF}_k^{\mathrm{adj}}(m) = \frac{\mathrm{WNF}_k(m) - k/(n-1)}{1 - k/(n-1)}, \qquad \mathrm{WNF}_k(m) = \frac{1}{n}\sum_{i=1}^{n} \frac{|\mathcal{N}_k^{\mathbf{x}}(i) \cap \mathcal{N}_k^{\mathbf{z},m}(i)|}{k}. \tag{6}$$

Here, $\mathcal{N}_k^{\mathbf{x}}(i)$ and $\mathcal{N}_k^{\mathbf{z},m}(i)$ denote the within-cell-type k-nearest-neighbor sets of cell i in expression and embedding space, respectively, and n is the number of cells of that type. A value of zero corresponds to the expected overlap under random neighbor assignment. We also assessed continuous within-type variation by using embeddings to predict major expression principal components. The two measurements quantify local and continuous state information after conditioning on cell-type identity.

Residual state structure. To impose a stricter anti-shortcut test, we removed cell-type centroid information symmetrically from both embedding and expression spaces:

$$\tilde{\mathbf{z}}_i^{(m)} = \mathbf{z}_i^{(m)} - \boldsymbol{\mu}_{c_i}^{(m)}, \qquad \tilde{\mathbf{x}}_i = \mathbf{x}_i - \boldsymbol{\mu}_{c_i}^{\mathbf{x}}. \tag{7}$$

Here, $\boldsymbol{\mu}_{c_i}^{(m)}$ and $\boldsymbol{\mu}_{c_i}^{\mathbf{x}}$ denote the corresponding cell-type centroids. Neighborhood fidelity and expression-PC predictability were then recomputed in the residual spaces. Preservation of local structure after residualization, particularly when exceeding both permutation and random-projection nulls, provides evidence that the embedding captures within-type state variation beyond centroid-level cell-type effects.

Gene-relation preservation. We next examined whether cell embeddings preserve gene–gene relationships that generalize to held-out cells. Gene loading vectors were estimated from the association between gene expression and embedding dimensions using training cells only. For genes g and h, the embedding-derived relation was defined as

$$C_{gh}^{(m)} = \cos\left(\boldsymbol{\ell}_g^{(m)}, \boldsymbol{\ell}_h^{(m)}\right), \tag{8}$$

where $\boldsymbol{\ell}_g^{(m)}$ denotes the embedding-derived loading vector of gene g. The resulting relation matrix was compared with the observed gene–gene co-expression matrix calculated exclusively from held-out cells. Gene-relation preservation was summarized as

$$S_{\text{gene}}(m) = \text{corr}\left[\text{vec}_{\triangle}\left(C^{\text{obs}}\right), \text{vec}_{\triangle}\left(C^{(m)}\right)\right]. \tag{9}$$

Higher values therefore indicate closer agreement between embedding-derived gene relationships and independently observed co-expression structure.

Perturbation direction consistency. We further tested whether perturbation responses exhibit biologically structured directional geometry rather than merely forming separable condition clusters. For each single-gene perturbation c, the embedding-space perturbation shift relative to control was defined as

$$\mathbf{v}_c^{(m)} = \mathbb{E}\left[\mathbf{z}^{(m)} \mid c\right] - \mathbb{E}\left[\mathbf{z}^{(m)} \mid \text{ctrl}\right]. \tag{10}$$

Pathway consistency was quantified by comparing cosine similarities between perturbation vectors belonging to the same pathway with those belonging to different pathways. Positive pathway-level separation indicates that perturbations affecting related biological programs induce more similar directions in embedding space. Significance was evaluated against gene-label permutation nulls with false-discovery-rate correction. For visualization, corrected values are transformed as $-\log_{10}(q)$; the significance threshold $q = 0.05$ therefore appears at $-\log_{10}(0.05) = 1.301$.

Cross-model agreement. Finally, we examined whether independently trained models converge towards a shared embedding-derived gene-relation geometry. For a selected gene set G, standardized gene expression X_G and model embedding $Z^{(m)}$ were used to obtain the gene–embedding loading matrix

$$L^{(m)} = \frac{X_G^\top Z^{(m)}}{n-1}, \qquad S^{(m)} = \tilde{L}^{(m)} \tilde{L}^{(m)\top}, \tag{11}$$

where $\tilde{L}^{(m)}$ contains ℓ_2-normalized gene loading vectors and $S^{(m)}$ represents the resulting model-specific gene-relation matrix. For each dataset, cross-model agreement was defined as the mean Pearson correlation between the upper-triangular entries of $S^{(m)}$ over all $\binom{10}{2} = 45$ model pairs, averaged across gene-set repeats. The resulting dataset-level consensus was compared with cell-shuffling, gene-shuffling, and dimension-matched random-projection nulls; it was not summarized by first computing separate model-level scores.

Although the observed cross-model agreement consistently exceeded the cell- and gene-shuffling nulls, it did not exceed the dimension-matched random-projection null. We therefore interpret this analysis conservatively as evidence of cross-model agreement in embedding-derived gene geometry, rather than evidence for enriched shared regulatory structure.

The six analyses address different levels of representational structure. Perturbation decoding, within-type neighborhood preservation and residual-state analysis test cellular variation beyond coarse cell-type identity. Gene-relation preservation and perturbation-direction consistency test gene- and pathway-level organization, while cross-model agreement evaluates convergence beyond the underlying expression geometry. Supplementary Note 3 describes dataset selection, preprocessing, cross-validation, null models, permutation tests and multiple-testing correction.

Hyperparameters

The downstream evaluation was implemented using PyTorch 2.1 and run on NVIDIA V100 GPUs with 32 GB memory. Unless otherwise specified, task-specific heads were optimized with Adam using a learning rate of 1×10^{-3} and a batch size of 32. Reconstruction and perturbation heads were trained for at most 500 epochs with validation-based early stopping; annotation heads were trained for 500 epochs. Annotation, reconstruction and perturbation experiments were repeated with five random seeds, and the reported task-level values are means across those replicates.

The implementation uses a common PyTorch interface for embedding extraction, downstream evaluation and visualization, with task-specific modules for perturbation prediction and cross-dataset comparison.

Statistics and Reproducibility

No new biological samples were collected; dataset sizes after benchmark-specific processing are reported in Supplementary Table 1. Annotation, reconstruction and perturbation-prediction experiments used random seeds 41–45 and are reported as arithmetic means across the five computational replicates. Clustering and batch correction are deterministic evaluations of the stored representations under the reported pipeline and are not five-seed estimates. Unless an explicit statistical test is stated, model rankings describe observed values or means and do not constitute inferential tests of superiority.

Representation Tests 1–3 and 5 use one-sided empirical permutation tests with 200 null draws and a plus-one correction. Resulting p values are adjusted by the Benjamini–Hochberg procedure within the comparison family specified for each test, with $q \leq 0.05$ used as the reporting threshold. Test 4 reports correlations as effect sizes across ten gene-set repeats and does not assign inferential significance. Test 6 uses ten draws per null family, giving a minimum attainable one-sided empirical p value of 1/11. Supplementary Note 3 specifies the test-specific samples, cross-validation folds, null hypotheses, comparison families and replicate definitions.

Data Availability

A study record for the processed data is available from the Broad Institute Single Cell Portal under accession SCP3702. The single-cell embeddings generated and used in this study are available from Hugging Face at cellbench-embeddings.

Code Availability

The benchmark was implemented in PyTorch. Source code, experiment configurations and analysis scripts are available at https://github.com/sky-Yongjie-Xu/2026-CellBench.

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Author Contributions

Z.Z. and F.H. proposed the study. Y.X., Y.L. and Z.Z. developed the method. Y.L., P.C. and H.T. collected the datasets. Y.X., Y.L. and Y.Y. designed and conducted the experiments, analysed the results and wrote the manuscript. All authors discussed the results, revised the manuscript and approved the final version.

Competing Interests

The authors declare no competing interests.